

Transforming Healthcare Operations with AI- Driven Insights

MedCore AI Operations Intelligence Dashboard



July 9th, 2026

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Executive Summary

An AI powered operational intelligence dashboard that helps MedCore Health work smarter, reduce administrative burden and make proactive decisions



CHALLENGE

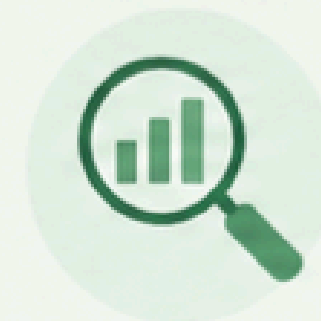
- Manual workflows and spreadsheets
- Repetitive administrative tasks
- Reactive decision making



OUR SOLUTION

AI Operations Dashboard

An AI-powered decision support tool that analyzes data, predicts risk, and recommends opportunities.



WHAT IT DOES

- Identifies operational inefficiencies
- Predicts departmental risk
- Recommends high-impact opportunities



BUSINESS IMPACT

- Reduces administrative burden
- Improves efficiency
- Enables proactive, data-driven decisions



Empowering MedCore Health to deliver **better care** through **smarter operations**.



Business Problem



Current State

MedCore relies on Manual workflows and repetitive administrative tasks to manage operations



Uneven Efficiency
Operational burden and risks vary significantly across departments.



High-Volume Admin Tasks
Tasks like Excel reporting, scheduling, data entry, patient follow-up, and claims reviews drive workload.



Excel Reporting Impact
Excel reporting is the largest contributor to operational burden across departments.



THE OPPORTUNITY



AI decision support tools can identify **high-risk departments**, prioritize areas needing attention, and recommend targeted improvements—before inefficiencies impact performance.

FROM REACTIVE TO PROACTIVE



A centralized system can help MedCore monitor patterns, reduce manual workload, and **address challenges before they occur.**



Data & Analytical Approach

Departments

Insight into budgets, wait times, manual workload, and operational risk.

Appointments

Shows number of wait days, no-show rate, and cancellations

Manual Workflow

Time logs of tasks, and shows which tasks are time consuming.

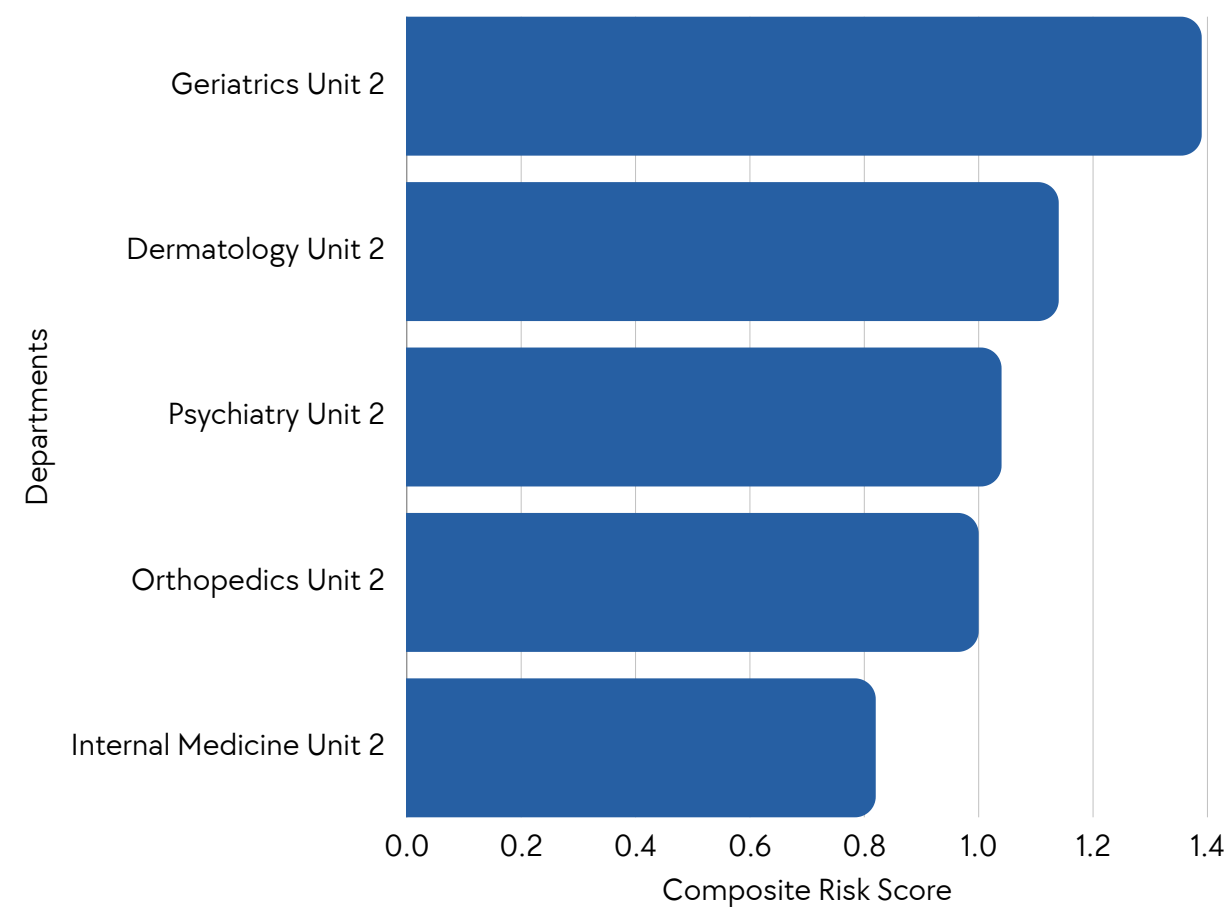
Composite Risk Score

Standardized z-score per department that combines five operational metrics into one number.

- **Total Manual Minutes** Staff and admin burden
- **Average Overbooking** Scheduling constraints
- **Average Wait Days** Delays experienced by patients
- **No Show-Rate** Gaps in patient follow-through
- **Error Rate** Gaps in quality and documentation

Composite Risk Score

Top 5 Departments with Highest Risk Score



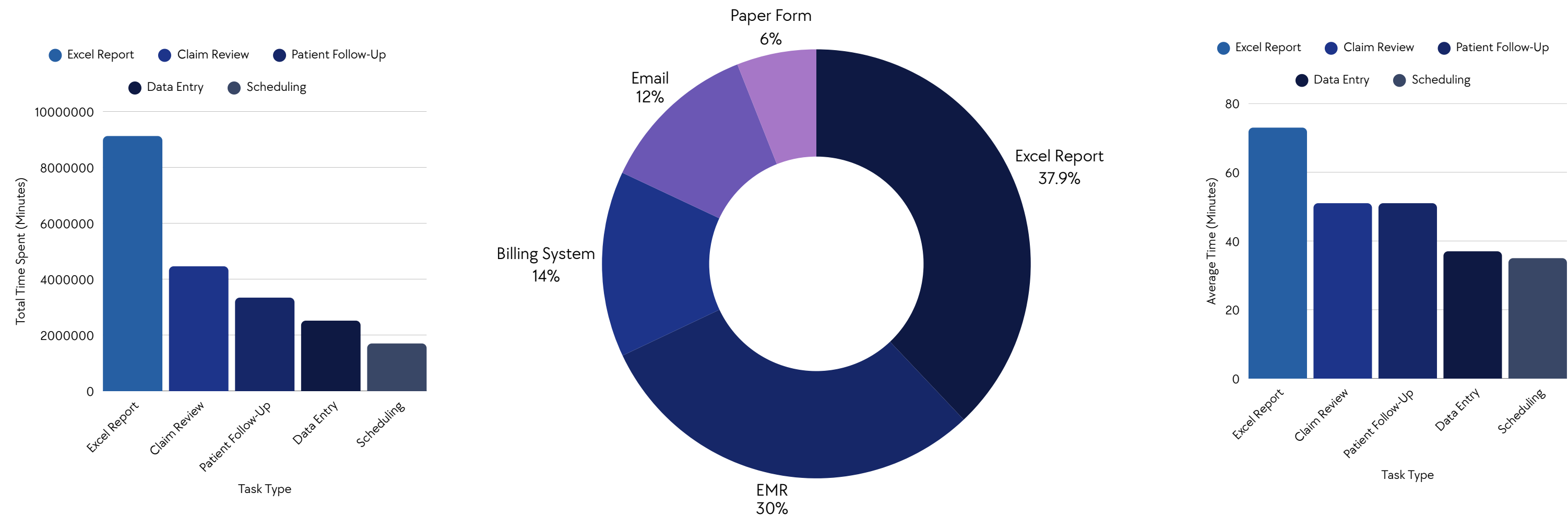
Top 5 Departments most at risk for operational inefficiencies

- Positive risk score indicates how elevated the risk is
- There is no single worst factor, instead they are elevated across all five simultaneously
- Pilot study target list



Key Findings

Largest Administrative Burden = Excel Reporting



- Excel reports were the most common task (124,552 uses).
- Most time-consuming task – **9.1 million minutes** and **73.2 minute average completion time**
- Excel was the most frequently used system (188,773 uses, 37.9% share).
- Large contributor to department’s risk score





Key Findings

Operational Inefficiencies: Departments Vary Significantly



0.71-3.18
manual
workload multiplier

Manual workload multipliers compare manual effort to the standard average.

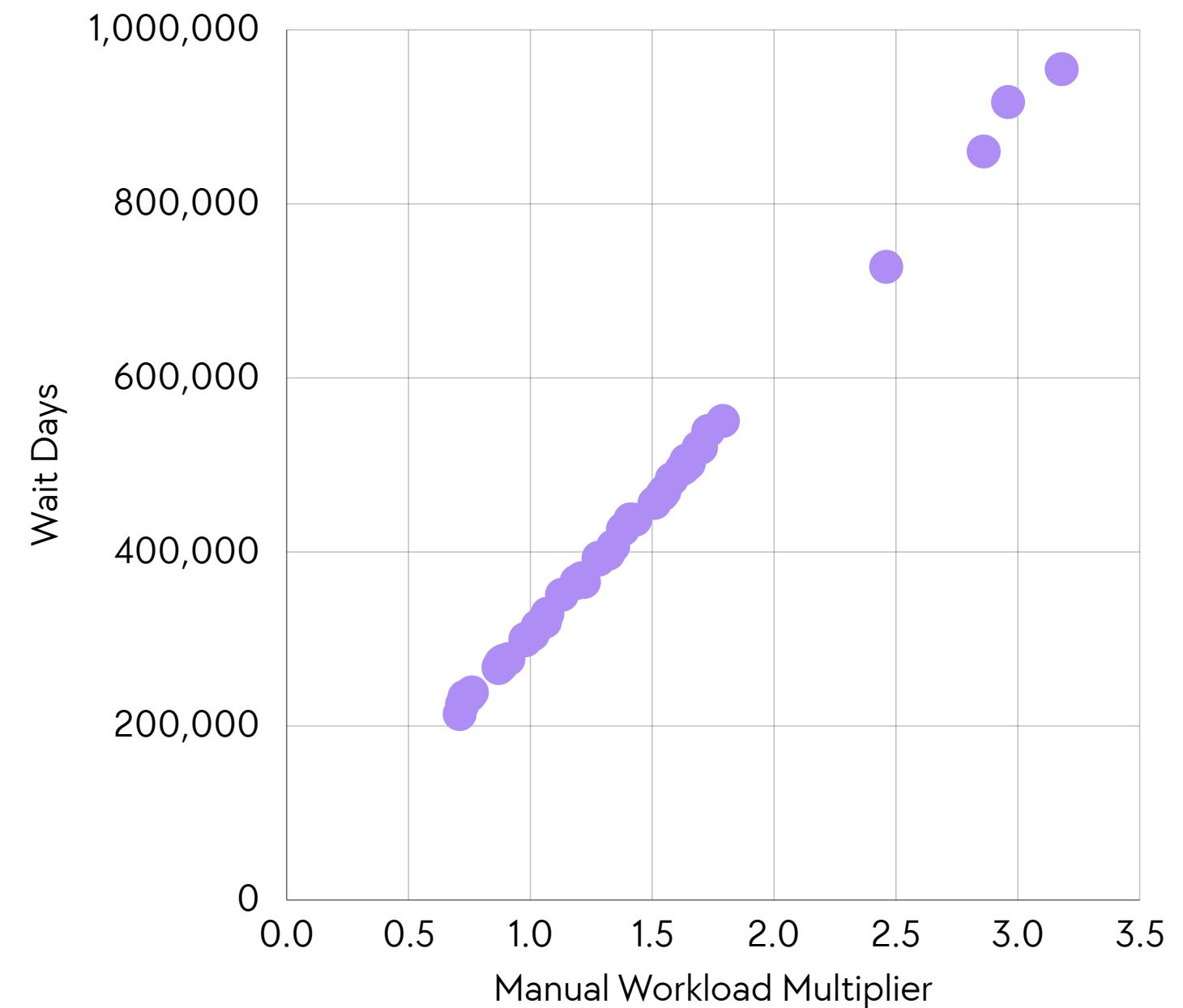
3-43
wait
days

Large department disparity in wait days.

$r \approx 1.00$
near-perfect relationship

Higher manual workload was a significant predictor of baseline and patient wait times.

Conclusion: Departments designed around manual or administrative processes were less efficient.



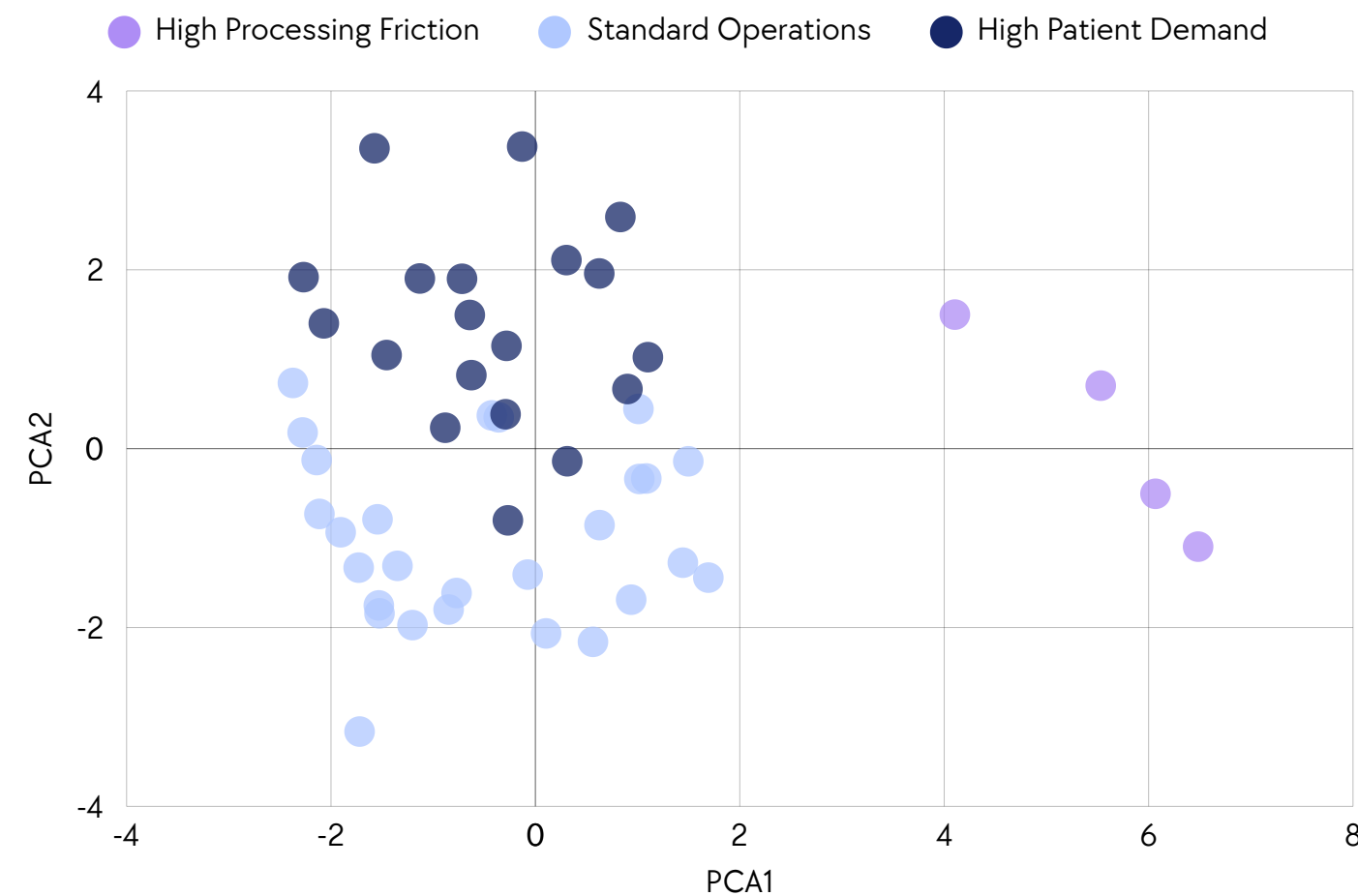
AI Approach

CORE NEEDS

Identify major workflow inefficiencies

Implement AI-powered productivity & consistency

Identify opportunities for automation



Risk Prediction Model

Output: Predicted Risk Score

Business Impact: Identified inefficiencies, intelligent decision-making

Explains 83.4% of the variation in department operational strain ($R^2 = 0.834$).

Average prediction error is 0.18 standardized units (MAE = 0.182), indicating accurate department-level risk estimation.

Department Clustering Model

Output: Intelligent Department Profiles

Business Impact: Identify system-wide patterns, establish best-practices



Proposed Solution: Operations Dashboard

Dashboard Business Impact



Accessible Department Summary Data

- Aggregated, department-level data highlighting challenges and inefficiencies of value to executives.



Intelligent Recommendations

- Inform management of possible AI automation solutions, improving decision making.



Department Clustering Models

- Allow decisions to be generalized to similar departments, saving management time.



Implementation Considerations & Future Opportunities

Key Rollout Considerations



Systems Integration

Pull live metrics directly from existing clinical databases to eliminate manual data transfers.



Safety

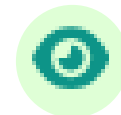
Hosting must remain highly secure and localized to protect sensitive medical and administrative data.



Staff Training

Train relevant staff on how and when to use the dashboard

Future Scaling Strategy



Monitor Success of Recommendations

Track AI recommendations and measure their effectiveness once implemented.



Cross-Department Rollout

After solving issues in highest priority departments, take successful solutions and apply to other departments as needed.



Predictive Prevention

Move from flagging active problems to predicting bottlenecks in advance.

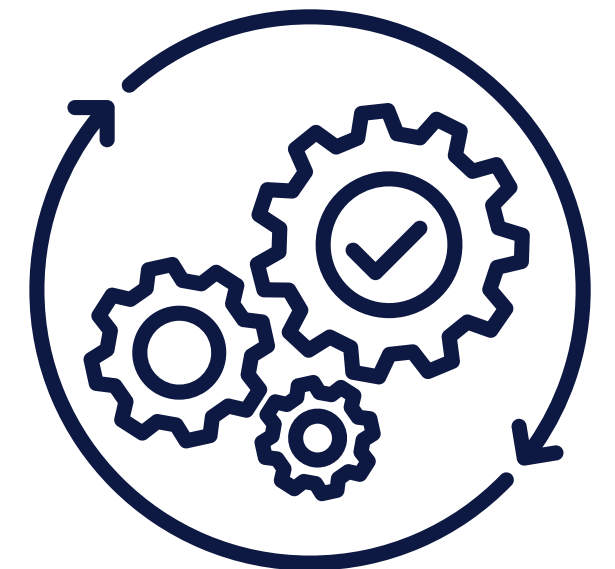
Closing Remarks

Medcore's Vulnerability

- Reliance on manual workflows and Excel-based systems and tasks.
- Highly correlated with inefficiencies like wait time.

AI Solution

- Our **MedCore AI Operations Dashboard** calculates a Composite Risk Score.
- Pinpoints where automation will yield the highest financial and operational return





**Thank
You!**

